AGENTIC AI USE CASE COMPLEXITY FRAMEWORK

Executive Summary: Agentic AI Use Case Complexity Framework

This framework categorizes agentic AI implementations across four complexity levels based on number of agents, orchestration complexity, data volume/variety, and UI requirements. Each level represents exponentially increasing technical sophistication and resource requirements.

Table

Complexity Level	Number of Agents	Development Time	Team Size	Monthly Cost	Total Project Cost	Key Characteristics
Small	1-2 agents	2-4 weeks	1-2 developers	$100-500	$5K-10K	Single-purpose automation; simple linear workflows; <1GB/day data; minimal UI; suitable for POC/MVP
Medium	3-5 agents	2-3 months	3-5 developers	$1K-5K	$10K-100K	Multi-step workflows with branching; 1-10GB/day data; moderate UI with visualizations; departmental solutions
Large	6-15 agents	6-12 months	8-15 developers	$10K-50K	$100K-1M	Complex parallel workflows; 10-100GB/day data; advanced dashboards; enterprise-wide, business-critical functions
Complex	15+ agents	12-24+ months	20-50+ developers	$50K-500K+	$1M+	Autonomous platforms; 100GB-1TB+/day data; multi-tenant enterprise UI; mission-critical 24/7 operations with SRE teams

Key Insights

Progression Path: Organizations should start with Small complexity to prove value, then scale progressively through Medium and Large before attempting Complex implementations. Complexity grows non-linearly—a Complex use case is exponentially more challenging than a simple chatbot.

Cost Drivers: The primary cost factors include agent count, data processing requirements, orchestration sophistication, and operational maintenance. Complex systems require dedicated SRE teams and 24/7 monitoring, significantly increasing ongoing costs.

Decision Criteria: Choose complexity level based on user base size, business criticality, budget constraints, and organizational readiness. Small suits <100 users and non-critical functions, while Complex serves 10,000+ users in mission-critical scenarios.

Details of approach

The current Accenture measures to define the sizing for t-shirt is unrealistic. I suggest a framework for categorizing agentic AI use cases by complexity levels based on the dimensions: number of agents, orchestration complexity, data volume/variety/processing, and UI requirements.

Complexity Dimensions

This framework categorizes agentic AI use cases across four complexity levels (Small, Medium, Large, Complex) based on five key dimensions:

1. Number of Agents: Single agent vs. multi-agent systems
2. Orchestration Complexity: Workflow patterns and coordination requirements
3. Data Volume: Amount of data processed
4. Data Variety: Types and sources of data
5. UI Requirements: User interface sophistication

SMALL COMPLEXITY USE CASES

Characteristics

Number of Agents: 1-2 agents
- Single agent with optional fallback/backup agent
- No inter-agent communication required
- Independent execution

Orchestration Complexity: Simple linear workflows
- Sequential task execution
- Minimal branching logic
- No parallel processing
- Single-step or 2-3 step workflows
- No human-in-the-loop requirements

Data Volume: Low
- <1GB per day
- <10K records per execution
- Batch processing acceptable
- No real-time requirements

Data Variety: Limited
- 1-2 data sources
- Single data format (e.g., text only)
- Structured data only
- No external API integrations

UI Requirements: Minimal
- Simple chat interface
- Basic input/output display
- No visualization needs
- No real-time updates

Example Use Cases

1. FAQ Chatbot
- Single agent answering common questions
- Knowledge base: Company documentation
- Data: Text-based FAQs (<100MB)
- UI: Simple chat widget
- Orchestration: Direct question → answer

2. Email Classifier
- Single agent categorizing incoming emails
- Data: Email text and metadata
- Volume: <5K emails/day
- UI: Email client integration (labels)
- Orchestration: Email → classify → tag

3. Simple Document Summarizer
- Single agent summarizing documents
- Data: PDF/Word documents
- Volume: <100 documents/day
- UI: Upload → summary display
- Orchestration: Upload → extract → summarize

4. Basic Sentiment Analysis
- Single agent analysing customer feedback
- Data: Text reviews/comments
- Volume: <1K reviews/day
- UI: Dashboard with sentiment scores
- Orchestration: Text → analyse → score

5. Product Recommendation (Simple)
- Single agent suggesting products
- Data: User profile + product catalogue
- Volume: <10K products
- UI: Product cards display
- Orchestration: User query → match → recommend

Implementation Characteristics

Technology Stack:
- AgentCore (Bedrock) with single agent
- No Strands orchestration needed
- Simple Lambda functions
- DynamoDB for state (if needed)
- Basic CloudWatch logging

Development Time: 2-4 weeks
Team Size: 1-2 developers
Cost: $100-500/month
Maintenance: Low (monthly reviews)

MEDIUM COMPLEXITY USE CASES

Characteristics

Number of Agents: 3-5 agents
- Multiple specialized agents
- Basic inter-agent communication
- Sequential or simple parallel execution
- Agent handoffs required

Orchestration Complexity: Multi-step workflows with branching
- 4-10 step workflows
- Conditional branching (if/then logic)
- Limited parallel processing
- Basic error handling and retries
- Optional human-in-the-loop for exceptions

Data Volume: Moderate
- 1-10GB per day
- 10K-100K records per execution
- Near real-time processing (minutes)
- Some streaming requirements

Data Variety: Moderate
- 3-5 data sources
- 2-3 data formats (text, structured data, images)
- Mix of structured and semi-structured data
- 2-3 external API integrations

UI Requirements: Moderate
- Multi-page application
- Basic visualizations (charts, graphs)
- Real-time status updates
- User input forms with validation

Example Use Cases

1. Customer Support Ticket Routing

- Agents: Classifier, Router, Responder, Escalation, Quality Checker
- Data: Tickets (text), customer history, knowledge base
- Volume: 10K-50K tickets/day
- UI: Support dashboard with ticket status
- Orchestration: Ticket → classify → route → respond → quality check

2. Content Moderation System
- Agents: Text Analyzer, Image Analyzer, Policy Checker, Decision Maker, Appeal Handler
- Data: User-generated content (text, images)
- Volume: 50K-100K items/day
- UI: Moderation queue with review interface
- Orchestration: Content → analyse → check policy → decide → log

3. Invoice Processing
- Agents: OCR Extractor, Data Validator, Matcher, Approver, Exception Handler
- Data: PDF invoices, purchase orders, vendor database
- Volume: 5K-20K invoices/day
- UI: Invoice review dashboard with approval workflow
- Orchestration: Upload → extract → validate → match → approve

4. Lead Qualification System
- Agents: Data Enricher, Scorer, Segmenter, Recommender, Follow-up Scheduler
- Data: Lead forms, CRM data, external enrichment APIs
- Volume: 10K-30K leads/day
- UI: Sales dashboard with lead scores and recommendations
- Orchestration: Lead → enrich → score → segment → recommend → schedule

5. Anomaly Detection & Alerting (Telco)
- Agents: Data Collector, Anomaly Detector, Root Cause Analyzer, Alert Generator, Ticket Creator
- Data: Network metrics (time-series), logs, topology
- Volume: 1M metrics/hour
- UI: Real-time monitoring dashboard with alerts
- Orchestration: Metrics → detect → analyse → alert → create ticket

Implementation Characteristics

Technology Stack:
- AgentCore (Bedrock) with multiple agents
- Strands for workflow orchestration
- LangGraph or AgentCore for state management
- MSK/Kafka for event streaming
- Timestream for time-series data
- OpenSearch for search/retrieval
- Grafana for monitoring

Development Time: 2-3 months
Team Size: 3-5 developers
Cost: $1K-5K/month
Maintenance: Moderate (weekly reviews)

LARGE COMPLEXITY USE CASES

Characteristics

Number of Agents: 6-15 agents
- Multiple specialized agent teams
- Complex inter-agent communication
- Parallel execution with synchronization
- Dynamic agent selection based on context
- Agent collaboration and negotiation

Orchestration Complexity: Complex workflows with multiple paths
- 10-30 step workflows
- Multiple conditional branches
- Extensive parallel processing
- Advanced error handling with compensation logic
- Human-in-the-loop at multiple decision points
- Workflow versioning and A/B testing

Data Volume: High
- 10-100GB per day
- 100K-1M records per execution
- Real-time processing (seconds)
- Continuous streaming

Data Variety: High
- 6-10 data sources
- 4-5 data formats (text, structured, images, audio, video)
- Mix of structured, semi-structured, and unstructured data
- 5-10 external API integrations
- Real-time data feeds

UI Requirements: Advanced
- Multi-tenant application
- Advanced visualizations (interactive dashboards)
- Real-time collaboration features
- Mobile and web interfaces
- Customizable workflows and views

Example Use Cases

1. Intelligent Manufacturing Operations Center
- Agents:
  - Data Collectors (3): Metrics, Logs, Topology
  - Analyzers (4): Anomaly, Root Cause, Impact, Prediction
  - Responders (3): Auto-remediation, Ticket Creation, Escalation
  - Coordinators (2): Workflow Manager, Human-in-Loop
  - Reporters (2): Dashboard, Notification
- Data: Network telemetry (100GB/day), logs, topology, tickets, runbooks
- Volume: 10M metrics/hour, 1M log lines/hour
- UI: Real-time dashboard with drill-down, mobile alerts
- Orchestration: Multi-stage detection → analysis → remediation → reporting

2. Fraud Detection & Prevention
- Agents:
  - Monitors (3): Transaction, Behaviour, Device
  - Analysers (4): Pattern Matcher, ML Scorer, Rule Engine, Graph Analyzer
  - Decision Makers (3): Risk Assessor, Action Recommender, Case Manager
  - Responders (2): Block/Allow, Alert Generator
  - Investigators (2): Case Builder, Evidence Collector
- Data: Transactions, user behaviour, device fingerprints, historical fraud cases
- Volume: 1M transactions/hour
- UI: Fraud analyst dashboard with case management
- Orchestration: Real-time transaction → multi-model analysis → decision → action

3. Personalized Learning Platform
- Agents:
  - Assessors (3): Skill Evaluator, Knowledge Tester, Progress Tracker
  - Recommenders (4): Content Suggester, Path Planner, Peer Matcher, Mentor Finder
  - Content Creators (3): Quiz Generator, Summary Creator, Exercise Builder
  - Tutors (3): Q&A Bot, Hint Provider, Explanation Generator
  - Coordinators (2): Learning Path Manager, Engagement Monitor
- Data: Student profiles, learning content, assessments, interactions
- Volume: 100K students, 1M interactions/day
- UI: Student portal, instructor dashboard, mobile app
- Orchestration: Continuous assessment → personalized recommendations → content delivery

4. Supply Chain Optimization
- Agents:
  - Forecasters (3): Demand, Supply, Risk
  - Optimizers (4): Inventory, Routing, Scheduling, Sourcing
  - Monitors (3): Shipment Tracker, Quality Inspector, Compliance Checker
  - Responders (3): Order Manager, Exception Handler, Supplier Communicator

  - Analysts (2): Performance Reporter, Cost Analyzer
- Data: Orders, inventory, shipments, supplier data, market data
- Volume: 100K orders/day, 10K shipments/day
- UI: Supply chain control tower with real-time visibility
- Orchestration: Continuous forecasting → optimization → execution → monitoring

5. Automated Trading System
- Agents:
  - Data Collectors (3): Market Data, News, Social Sentiment
  - Analysers (4): Technical Analyzer, Fundamental Analyzer, Sentiment Analyzer, Risk Assessor
  - Strategists (3): Signal Generator, Portfolio Optimizer, Execution Planner
  - Traders (2): Order Executor, Position Manager
  - Monitors (3): Performance Tracker, Risk Monitor, Compliance Checker
- Data: Market data (tick-by-tick), news feeds, financial statements
- Volume: 1M ticks/second, 10K news articles/day
- UI: Trading dashboard with real-time P&L, risk metrics
- Orchestration: Continuous data ingestion → multi-model analysis → strategy → execution

Implementation Characteristics

Technology Stack:
- AgentCore (Bedrock) with agent teams
- Strands for complex workflow orchestration
- LangGraph for advanced state management
- MSK/Kafka for high-throughput streaming
- Timestream for time-series data
- Redshift for analytics
- Neptune for graph relationships
- OpenSearch for search/retrieval
- ElastiCache for caching
- DynamoDB for state management
- S3 for data lake
- Grafana + CloudWatch + X-Ray for observability

Development Time: 6-12 months
Team Size: 8-15 developers
Cost: $10K-50K/month
Maintenance: High (daily monitoring, weekly optimization)

COMPLEX COMPLEXITY USE CASES

Characteristics

Number of Agents: 15+ agents
- Large-scale multi-agent systems
- Hierarchical agent organizations
- Dynamic agent creation and termination
- Agent learning and adaptation
- Cross-domain agent collaboration
- Agent marketplaces or ecosystems

Orchestration Complexity: Highly complex, adaptive workflows
- 30+ step workflows
- Dynamic workflow generation
- Self-optimizing workflows
- Multi-level human-in-the-loop
- Workflow learning from outcomes
- Cross-workflow dependencies
- Real-time workflow adaptation

Data Volume: Very High
- 100GB-1TB+ per day
- 1M+ records per execution
- Sub-second real-time processing
- Massive parallel streaming

Data Variety: Very High
- 10+ data sources
- All data formats (text, structured, images, audio, video, sensor data)
- Structured, semi-structured, unstructured, and streaming data
- 10+ external API integrations
- Real-time data feeds from multiple domains
- Multi-modal data fusion

UI Requirements: Enterprise-grade
- Multi-tenant, multi-region application
- Advanced analytics and BI
- Real-time collaboration and communication
- Mobile, web, and API interfaces
- Customizable workflows, dashboards, and reports
- White-label capabilities
- Accessibility compliance

Example Use Cases

1. Autonomous Enterprise Operations Platform
- Agents (20+):

- IT Operations (5): Infrastructure Monitor, Incident Responder, Capacity Planner, Security Analyzer, Cost Optimizer
- Business Operations (5): Process Miner, Workflow Optimizer, Resource Allocator, Performance Tracker, Compliance Monitor
- Customer Operations (5): Support Orchestrator, Sentiment Analyzer, Churn Predictor, Upsell Recommender, Feedback Analyzer
- Data Operations (5): Data Quality Monitor, Pipeline Optimizer, Schema Evolver, Lineage Tracker, Governance Enforcer
- Coordinators (3): Master Orchestrator, Human-in-Loop Manager, Learning Coordinator
- Data: All enterprise systems (ERP, CRM, ITSM, logs, metrics, transactions)
- Volume: 1TB/day, 100M events/day
- UI: Unified operations center with role-based views
- Orchestration: Continuous monitoring → multi-domain analysis → coordinated actions → learning

2. Smart City Management System
- Agents (25+):
- Transportation (6): Traffic Monitor, Signal Optimizer, Route Planner, Parking Manager, Public Transit Coordinator, Incident Responder
- Utilities (5): Energy Optimizer, Water Monitor, Waste Manager, Grid Balancer, Outage Responder
- Public Safety (5): Crime Predictor, Emergency Dispatcher, Resource Allocator, Crowd Monitor, Disaster Coordinator
- Environment (4): Air Quality Monitor, Noise Analyzer, Weather Predictor, Pollution Tracker
- Services (3): Permit Processor, Complaint Handler, Service Scheduler
- Coordinators (2): City Operations Manager, Citizen Engagement Manager
- Data: IoT sensors (millions), cameras, social media, 311 calls, permits, incidents
- Volume: 10TB/day, 1B sensor readings/day
- UI: City operations dashboard, citizen mobile app, public displays
- Orchestration: Real-time sensor fusion → multi-domain optimization → coordinated response

3. Autonomous Clinical Decision Support System
- Agents (30+):
- Diagnostics (8): Symptom Analyzer, Lab Interpreter, Imaging Analyzer, Differential Diagnosis Generator, Risk Assessor, Genetic Analyzer, Drug Interaction Checker, Allergy Checker
- Treatment (6): Treatment Recommender, Dosage Calculator, Protocol Selector, Outcome Predictor, Side Effect Monitor, Adherence Tracker
- Monitoring (5): Vital Signs Monitor, Deterioration Detector, Complication Predictor, Recovery Tracker, Readmission Predictor
- Coordination (4): Care Team Coordinator, Appointment Scheduler, Referral Manager, Handoff Facilitator
- Documentation (3): Clinical Note Generator, Coding Assistant, Quality Measure Tracker
- Research (2): Clinical Trial Matcher, Evidence Synthesizer
- Oversight (2): Safety Monitor, Ethics Advisor
- Data: EHR, lab results, imaging, genomics, wearables, literature, clinical trials
- Volume: 100GB/day per hospital, multi-modal data
- UI: Clinician dashboard, patient portal, mobile app

- Orchestration: Continuous patient monitoring → multi-specialty analysis → coordinated care → outcome tracking

4. Global Supply Chain Resilience Platform
- Agents (35+):
  - Demand (5): Forecaster, Trend Analyzer, Seasonality Detector, Promotion Planner, Market Analyzer
  - Supply (6): Supplier Monitor, Capacity Tracker, Quality Assessor, Risk Evaluator, Alternative Sourcer, Contract Manager
  - Logistics (7): Route Optimizer, Carrier Selector, Shipment Tracker, Customs Navigator, Warehouse Manager, Last-Mile Coordinator, Returns Handler
  - Inventory (5): Stock Optimizer, Allocation Manager, Replenishment Planner, Obsolescence Detector, Safety Stock Calculator
  - Risk (5): Disruption Detector, Impact Assessor, Mitigation Planner, Scenario Simulator, Continuity Manager
  - Finance (4): Cost Analyzer, Payment Optimizer, Currency Manager, Working Capital Optimizer
  - Sustainability (3): Carbon Tracker, Circular Economy Advisor, ESG Reporter
- Data: Orders, shipments, inventory, supplier data, market data, weather, geopolitics, IoT
- Volume: 1TB/day, 10M transactions/day
- UI: Global control tower, supplier portal, customer portal
- Orchestration: End-to-end visibility → predictive analytics → proactive optimization → resilient execution

5. Autonomous Financial Services Platform
- Agents (40+):
  - Trading (8): Market Analyzer, Signal Generator, Strategy Optimizer, Order Executor, Position Manager, Risk Monitor, Performance Tracker, Compliance Checker
  - Risk Management (7): Credit Assessor, Market Risk Analyzer, Operational Risk Monitor, Liquidity Manager, Stress Tester, Scenario Analyzer, Capital Optimizer
  - Customer Service (6): Onboarding Assistant, Account Manager, Product Recommender, Complaint Handler, Fraud Detector, Churn Predictor
  - Operations (6): Transaction Processor, Reconciliation Manager, Settlement Coordinator, Reporting Generator, Audit Trail Manager, Exception Handler
  - Compliance (5): AML Monitor, KYC Verifier, Regulatory Reporter, Policy Enforcer, Sanctions Screener
  - Analytics (4): Customer Segmenter, Profitability Analyzer, Trend Forecaster, Benchmark Comparator
  - Innovation (4): Product Designer, Pricing Optimizer, Channel Analyzer, Partnership Evaluator
- Data: Transactions, market data, customer data, regulatory data, news, social media
- Volume: 10TB/day, 100M transactions/day
- UI: Trading platform, customer portal, advisor dashboard, compliance dashboard
- Orchestration: Real-time transaction processing → multi-model risk analysis → regulatory compliance → customer engagement

Implementation Characteristics

Technology Stack:
- AgentCore (Bedrock) with large-scale agent orchestration
- Strands for enterprise workflow orchestration
- LangGraph for complex state management
- MSK/Kafka for massive-scale streaming
- Timestream for time-series data
- Redshift for enterprise analytics
- Neptune for complex graph relationships
- OpenSearch for enterprise search
- ElastiCache for distributed caching
- DynamoDB for global state management
- S3 for data lake
- Grafana + CloudWatch + X-Ray for comprehensive observability
- AWS Step Functions for workflow coordination
- SageMaker for custom ML models
- Lambda for serverless compute
- ECS/EKS for containerized workloads

Development Time: 12-24+ months
Team Size: 20-50+ developers, architects, data scientists
Cost: $50K-500K+/month
Maintenance: Very High (24/7 monitoring, continuous optimization, dedicated SRE team)

DECISION FRAMEWORK

When to Choose Each Complexity Level

Small:
- Proof of concept or MVP
- Single-purpose automation
- Limited user base (<100 users)
- Non-critical business function
- Budget constraints (<$10K total)

Medium:
- Production-ready departmental solution
- Multi-step business process
- Moderate user base (100-1,000 users)
- Important but not mission-critical
- Budget: $10K-100K total

Large:
- Enterprise-wide solution
- Complex business process

- Large user base (1,000-10,000 users)
- Business-critical function
- Budget: $100K-1M total

Complex:
- Strategic platform or ecosystem
- Multi-domain, cross-functional
- Very large user base (10,000+ users)
- Mission-critical, 24/7 operation
- Budget: $1M+ total

SCALING CONSIDERATIONS

Progression Path

Most organizations should follow this progression:

1. Start Small: Prove value with simple use case
2. Scale to Medium: Add complexity as you learn
3. Expand to Large: Enterprise deployment with governance
4. Evolve to Complex: Platform approach with ecosystem

Key Success Factors by Level

Small:
- Clear, narrow scope
- Fast iteration
- Simple metrics
- Minimal dependencies

Medium:
- Well-defined workflows
- Data quality focus
- User training
- Basic monitoring

Large:
- Robust architecture
- Comprehensive testing
- Change management
- Advanced observability
- FinOps discipline

Complex:
- Enterprise architecture
- Governance framework
- SRE practices
- Continuous optimization
- Innovation pipeline
- Ecosystem management

This framework provides a structured way to assess and categorize agentic AI use cases based on their complexity. The key insight is that complexity grows non-linearly across dimensions—a use case with 15+ agents, complex orchestration, high data volume, and enterprise UI requirements is exponentially more complex than a simple chatbot.

For an anomaly detection and auto-remediation use case, based on sub-minute detection with multiple data sources (metrics, logs, topology), this would likely fall into the large complexity category:
- 6-10 agents (collectors, detectors, analysers, remediators)
- Complex orchestration with parallel processing
- High data volume (millions of metrics/hour)
- Multiple data sources and formats
- Real-time dashboard requirements